Prophet Model Implementation for Digital Payment Volume Forecasting with Holiday and Event Effects

Applied Time Series Analysis (ATSA): Group 2

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Overview

Problem. UPI processes hundreds of millions of payments a day, and daily volume is not smooth. It rises on weekends and paydays, ramps up before Diwali, dips on national holidays and jumps during e-commerce sales. Daily forecasts are needed for server capacity, fraud-monitoring baselines and settlement liquidity, and a forecast that ignores the calendar is wrong on exactly the busiest days.

What we do. We forecast daily UPI transaction volume with **Prophet** and test how much the forecast improves when Prophet is told about **holidays** (Diwali, Holi, Eid, Raksha Bandhan, national holidays) and **events** (an e-commerce sale, payday and month-end). We compare against a seasonal naïve baseline and **SARIMAX**.

Data. NPCI publishes UPI volume **monthly only**, so we built a **synthetic daily series** anchored to the real NPCI monthly totals (Jan 2022 – Aug 2026), with holiday and event effects of known size. Because the true effects are known, we can also check whether Prophet **recovers** them. *All daily values are synthetic; only the monthly totals are real.* Results therefore evaluate the modelling approach under our stated simulation assumptions, not the real causal effect of festivals on UPI. Details are in `DATA.md`.

Questions

1. Does adding holidays and events to Prophet improve forecast accuracy, and on which days?
2. Does Prophet recover the true size of each holiday and event effect?
3. How does Prophet compare with SARIMAX?

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1	Data
2	Exploratory data analysis
3	Models
4	Forecast evaluation

Section	Content
5	Holiday and event effects
6	Residual diagnostics
7	Final forecast
8	Conclusions

1. Data

1.1 Setup

In [1]:
```
import json, logging, warnings, itertools, time
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from joblib import Parallel, delayed

from prophet import Prophet
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.seasonal import STL
from statsmodels.tsa.stattools import adfuller, kpss, acf
from statsmodels.stats.diagnostic import acorr_ljungbox, het_arch
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf


def quiet():
    """Silence Stan/Prophet log messages (also called inside parallel
workers)."""
    warnings.filterwarnings("ignore")
    for name in ("cmdstanpy", "prophet", "prophet.plot"):
        logging.getLogger(name).disabled = True


quiet()
DATA, OUT = Path("data"), Path("results")
OUT.mkdir(exist_ok=True)
N_JOBS = -1          # run cross-validation folds on all CPU cores
RANDOM_SEED = 42
np.random.seed(RANDOM_SEED)

C_ACTUAL, C_MODEL, C_TRUTH, C_ALT, C_GRID = "#2a78d6", "#eb6834", "#52514e",
"#1baf7a", "#e4e3df"
plt.rcParams.update({
    "figure.dpi": 110, "savefig.dpi": 150, "savefig.bbox": "tight",
    "axes.spines.top": False, "axes.spines.right": False, "axes.grid": True,
    "grid.color": C_GRID, "grid.linewidth": 0.6, "axes.titlesize": 11,
"axes.titleweight": "bold",
    "axes.labelsize": 9, "xtick.labelsize": 8, "ytick.labelsize": 8,
```

```
    "legend.fontsize": 8, "legend.frameon": False, "lines.linewidth": 1.4,
})


def show(fig, name):
    fig.savefig(OUT / f"{name}.png")
    plt.show()
```

1.2 Load the data

- `upi_monthly.csv` : the real NPCI monthly totals, scraped from npci.org.in.
- `synthetic_daily.csv` : the daily series built from them:
  - `y` is daily volume in millions of transactions.
  - `true_effect` is the injected holiday and event effect on each day, which is our ground truth.
- `prophet_holidays.csv` : festival and holiday dates with their windows.
- `holidays.csv` : the gazetted national holiday list.
- `truth.json` : every injected parameter, including the sale dates.

How the daily series is built (short version):

- A smooth trend through NPCI's monthly averages.
- Multiplied by weekday, payday, holiday, sale and noise effects.
- Every month then sums exactly to the NPCI total, except five festive months.

The five festive months are not forced to match. Forcing them pushed their ordinary days about 5% below trend and hid half of the Diwali effect. We found this in the first version of our generator and fixed it (`DATA.md` §3.4).

In [2]:
```
df = pd.read_csv(DATA / "synthetic_daily.csv", parse_dates=["ds"])
monthly = pd.read_csv(DATA / "upi_monthly.csv", parse_dates=["month"])
festivals = pd.read_csv(DATA / "prophet_holidays.csv", parse_dates=["ds"])
national = pd.read_csv(DATA / "holidays.csv", parse_dates=["ds"])
truth = json.load(open(DATA / "truth.json"))

# Basic data integrity checks: fail early if the input series is not usable.
required_cols = {"ds", "y", "is_payday", "is_monthend", "true_effect"}
missing_cols = required_cols - set(df.columns)
if missing_cols:
    raise ValueError(f"synthetic_daily.csv is missing columns:
{sorted(missing_cols)}")
if df["ds"].duplicated().any():
    raise ValueError("Duplicate dates found in synthetic_daily.csv")
if not df["ds"].is_monotonic_increasing:
    df = df.sort_values("ds").reset_index(drop=True)
if df["y"].isna().any() or (df["y"] ⩽ 0).any():
    raise ValueError("Daily volume must be positive and non-missing for log-
scale models")
expected_dates = pd.date_range(df.ds.min(), df.ds.max(), freq="D")
if len(expected_dates) ≠ len(df) or not
pd.DatetimeIndex(df.ds).equals(expected_dates):
    raise ValueError("Daily series has missing or irregular dates")
print(f"{len(df)} days, {df.ds.min().date()} to {df.ds.max().date()}")
df[["ds", "y", "is_payday", "is_monthend", "true_effect"]].head()
```

```
1704 days, 2022-01-01 to 2026-08-31
```

Out[2]:

	ds	y	is_payday	is_monthend	true_effect
0	2022-01-01	155.413512	1	0	0.0
1	2022-01-02	142.543802	1	0	0.0
2	2022-01-03	149.680703	1	0	0.0
3	2022-01-04	150.994083	1	0	0.0
4	2022-01-05	150.568082	1	0	0.0

1.3 Holidays and events

Holidays are recurring public or cultural dates. Each one gets a **window** of days around it, because spending changes before and after the day itself:

Holiday	Window (days)	Injected effect
Diwali	−10 to +3	ramp up to +20% on Dhanteras and Diwali eve, then a dip
Holi	−2 to +1	+8% on the two days before, −10% on the day
Eid al-Fitr	−3 to 0	ramps up to +6%
Raksha Bandhan	−3 to 0	ramps up to +5%

Holiday	Window (days)	Injected effect
Republic Day, Independence Day, Gandhi Jayanti	0	−5%

Other gazetted holidays (Dussehra, Christmas, …) are also given to the model as single days. Their true effect is zero, so they test whether the model wrongly invents effects.

Events are non-holiday calendar effects:

- **E-commerce sale.** One 4–6 day sale per year, about 2½ weeks before Diwali, lifting volume by 8–12%. Sale dates are announced in advance by retailers, so they can be supplied to a forecast. The dates come from `truth.json`.
- **Payday.** Days 1–5 of each month, when salaries are credited: +6%.
- **Month-end.** The last 2 days of each month: +3%.

Each day is labelled for the evaluation: **holiday window**, **sale event**, or **ordinary**.

In [3]:
```
sale_rows = truth["hidden_sale_NOT_given_to_prophet"]   # the generator's
name for the sale; we now use it as an event
sale = pd.DataFrame({"holiday": "ecommerce_sale",
                     "ds": pd.to_datetime([s["start"] for s in sale_rows]),
                     "lower_window": 0,
                     "upper_window": [s["days"] - 1 for s in sale_rows]})
national_single = (national.assign(lower_window=0, upper_window=0)
                   [["holiday", "ds", "lower_window", "upper_window"]])
national_single =
national_single[~national_single.ds.isin(set(festivals.ds))]   # a window
replaces the same-date single day
HOLIDAYS = pd.concat([festivals, national_single], ignore_index=True)
HOLIDAYS_AND_EVENTS = pd.concat([HOLIDAYS, sale], ignore_index=True)


def window_days(table):
    days = set()
    for _, r in table.iterrows():
        days |= set(pd.date_range(r.ds +
pd.Timedelta(days=int(r.lower_window)),
                                  r.ds +
pd.Timedelta(days=int(r.upper_window))))
    return days


sale_days = window_days(sale)
df["day_type"] = np.select([df.ds.isin(sale_days),
df.ds.isin(window_days(festivals))],
                           ["sale event", "holiday window"], "ordinary")
DAY_TYPES = ["ordinary", "holiday window", "sale event"]
sale
```

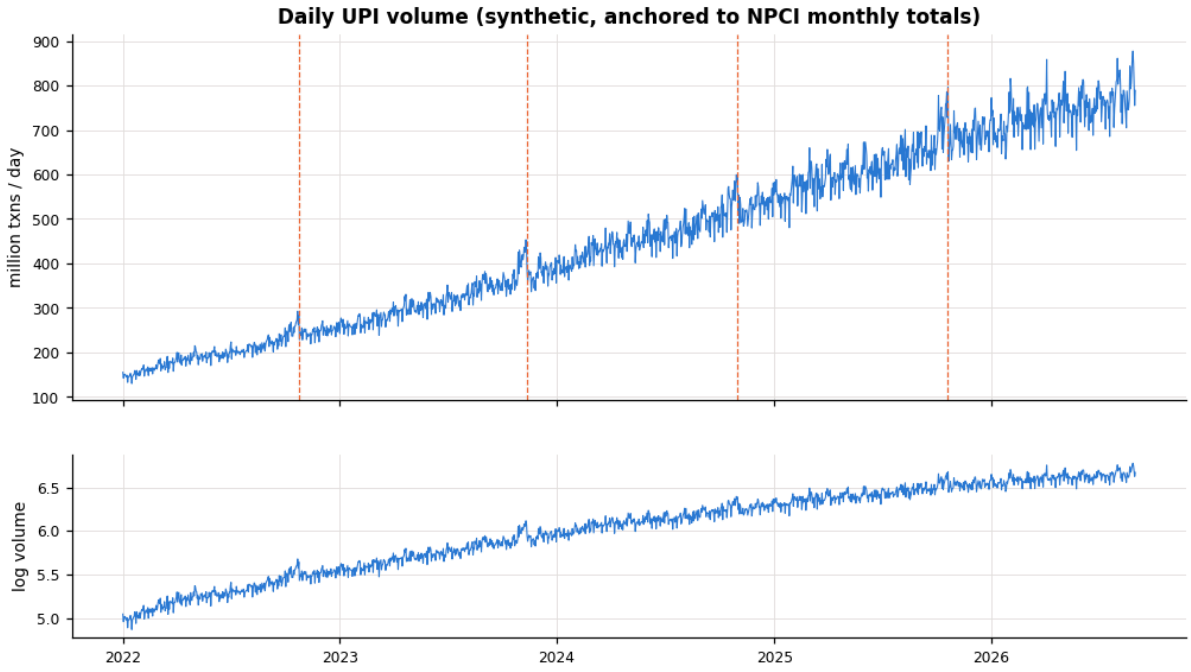
Out[3]:

	holiday	ds	lower_window	upper_window
0	ecommerce_sale	2022-10-07	0	4
1	ecommerce_sale	2023-10-28	0	3
2	ecommerce_sale	2024-10-15	0	3
3	ecommerce_sale	2025-10-04	0	5
4	ecommerce_sale	2026-10-21	0	4

2. Exploratory data analysis

2.1 The series

In [4]:
```
fig, ax = plt.subplots(2, 1, figsize=(11, 6), sharex=True, gridspec_kw=
{"height_ratios": [2, 1]})
ax[0].plot(df.ds, df.y, color=C_ACTUAL, lw=0.7)
ax[0].set_ylabel("million txns / day")
ax[0].set_title("Daily UPI volume (synthetic, anchored to NPCI monthly
totals)")
for d in festivals.query("holiday == 'diwali'").ds:
    if d ⩽ df.ds.max():
        ax[0].axvline(d, color=C_MODEL, lw=0.8, ls="--")
ax[1].plot(df.ds, np.log(df.y), color=C_ACTUAL, lw=0.7)
ax[1].set_ylabel("log volume")
show(fig, "2_1_series")
```

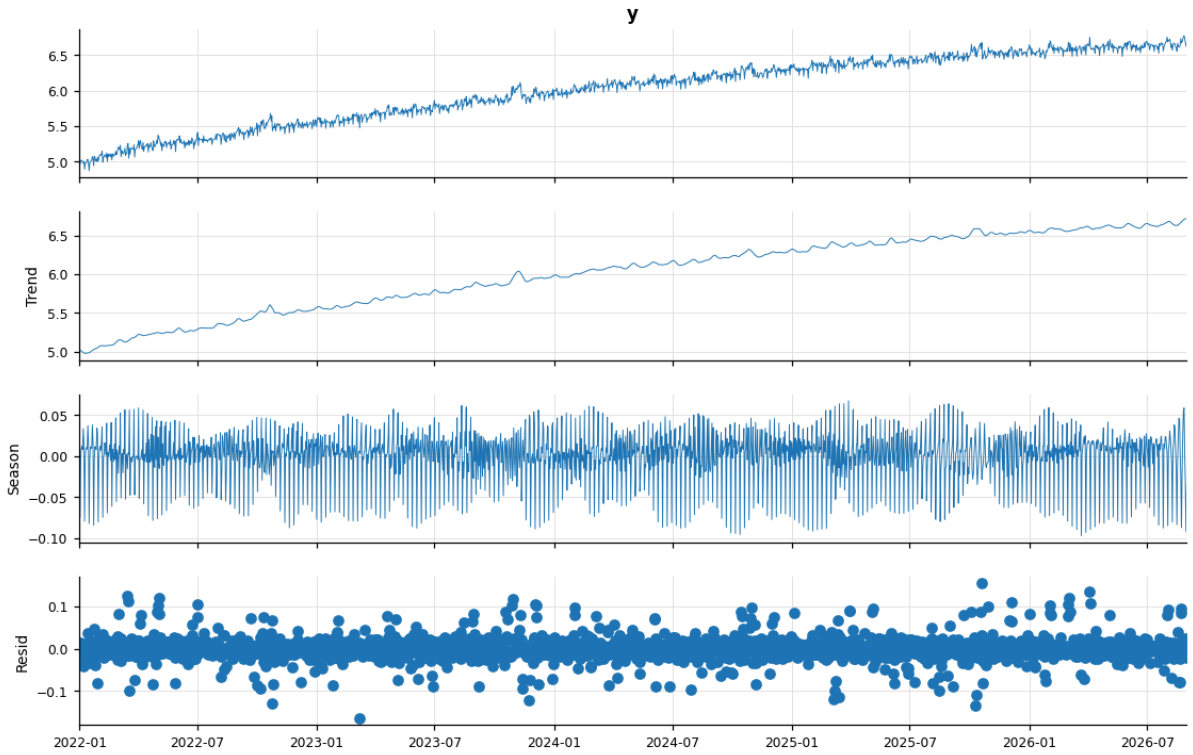


Reading:

- **Volume grows about fivefold** over the period, from about 150 to about 800 million transactions a day.
- **On the log scale the growth is close to linear,** gently slowing.
- **Festival seasons (dashed lines: Diwali) show visible bumps.**
- **The weekly up-and-down pattern grows with the level.** That is the sign of **multiplicative** seasonality.

2.2 STL decomposition (log scale, weekly period)

In [5]:
```
stl = STL(np.log(df.set_index("ds").y), period=7, robust=True).fit()
fig = stl.plot()
fig.set_size_inches(11, 7)
for a in fig.axes:
    for line in a.get_lines():
        line.set_linewidth(0.6)
show(fig, "2_2_stl")
```



Reading:

- **The weekly component is large and stable,** with Sunday about 7–9% below the weekly mean.
- **The festival seasons appear as bumps in the *trend* panel** each Oct–Nov. A two-week festival is long enough for STL to absorb it into the trend, so a plain decomposition cannot separate festivals from growth. That is why holidays have to be modelled explicitly.

2.3 Stationarity and autocorrelation

- **ADF test:** the null hypothesis is a unit root (non-stationary).
- **KPSS test:** the null hypothesis is stationarity.

A series counts as stationary only when both agree.

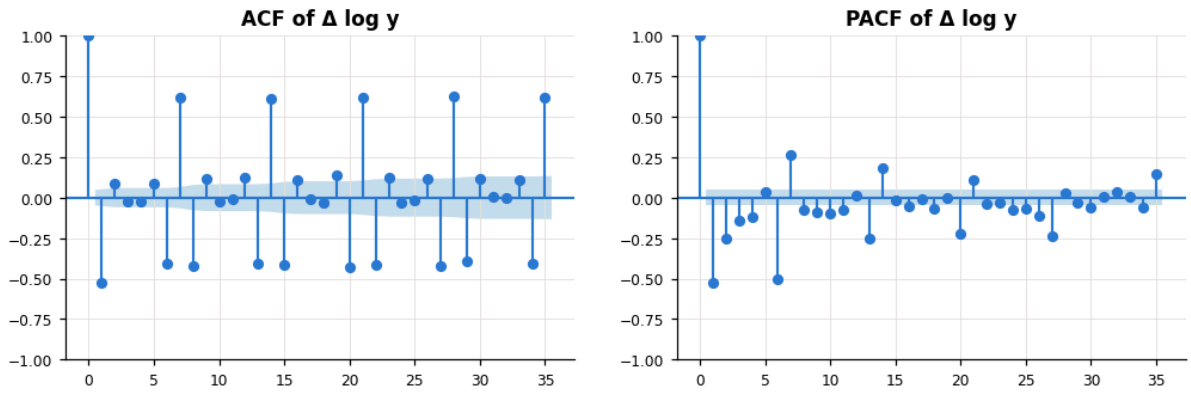
In [6]:
```
def stationarity(x, label):
    adf_p, kpss_p = adfuller(x, autolag="AIC")[1], kpss(x, regression="c",
nlags="auto")[1]
    return {"series": label, "ADF p": round(adf_p, 4), "KPSS p":
round(kpss_p, 4),
            "verdict": "stationary" if adf_p < 0.05 and kpss_p > 0.05 else
"non-stationary"}


ly = np.log(df.y.values)
pd.DataFrame([stationarity(ly, "log y"), stationarity(np.diff(ly), "Δ log y
(first difference)")])
```

Out[6]:

	series	ADF p	KPSS p	verdict
0	log y	0.0739	0.01	non-stationary
1	Δ log y (first difference)	0.0000	0.10	stationary

In [7]:
```
fig, ax = plt.subplots(1, 2, figsize=(11, 3.2))
plot_acf(np.diff(ly), lags=35, ax=ax[0], color=C_ACTUAL, vlines_kwargs=
{"colors": C_ACTUAL})
plot_pacf(np.diff(ly), lags=35, ax=ax[1], method="ywm", color=C_ACTUAL,
vlines_kwargs={"colors": C_ACTUAL})
ax[0].set_title("ACF of Δ log y")
ax[1].set_title("PACF of Δ log y")
show(fig, "2_3_acf_pacf")
```



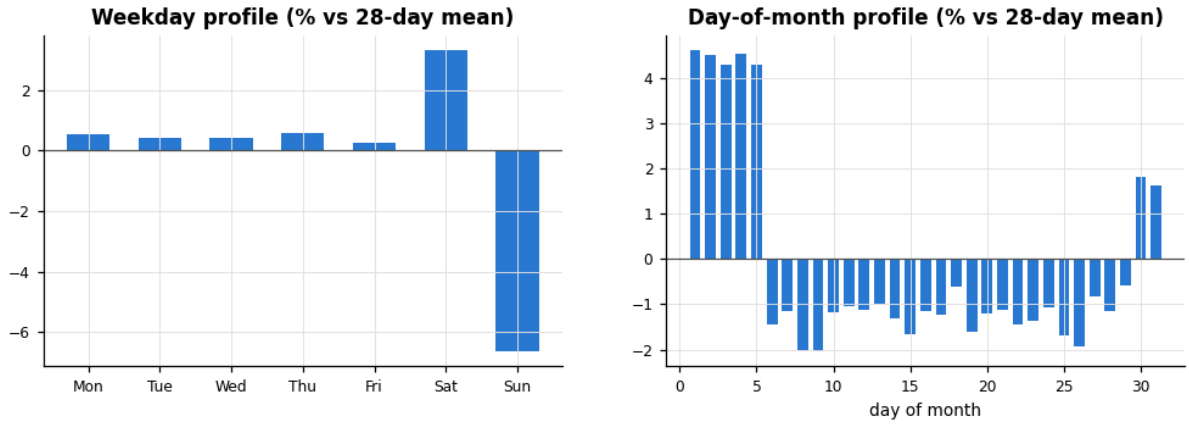
Reading:

- **Log volume is non-stationary; one difference makes it stationary.** So SARIMAX uses $d = 1$.
- **The ACF has strong spikes at lags 7, 14, 21, 28,** a weekly cycle, so the seasonal period is $s = 7$. The spikes barely decay, so the weekly pattern is stable. We model it with seasonal AR/MA terms ($D = 0$).
- **The large negative lag-1 value** points to low-order non-seasonal AR/MA terms.

2.4 Weekday and day-of-month profiles

Each day is divided by a centred 28-day average (to remove the trend). Holiday and sale days are left out, then the ratios are averaged.

In [8]:
```
tmp = df.set_index("ds").copy()
tmp["ratio"] = tmp.y / tmp.y.rolling(28, center=True).mean()
tmp = tmp[(tmp.day_type == "ordinary") & tmp.ratio.notna()]
fig, ax = plt.subplots(1, 2, figsize=(11, 3.2))
wd = tmp.groupby(tmp.index.dayofweek).ratio.mean()
dm = tmp.groupby(tmp.index.day).ratio.mean()
ax[0].bar(["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"], (wd - 1) * 100,
color=C_ACTUAL, width=0.6)
ax[0].set_title("Weekday profile (% vs 28-day mean)")
ax[1].bar(dm.index, (dm - 1) * 100, color=C_ACTUAL, width=0.7)
ax[1].set_title("Day-of-month profile (% vs 28-day mean)")
ax[1].set_xlabel("day of month")
for a in ax:
    a.axhline(0, color=C_TRUTH, lw=0.8)
show(fig, "2_4_calendar_profiles")
```



Reading:

- **Weekdays:** Saturday is about 3% above average and Sunday about 6.5% below.
- **Paydays:** days 1–5 are about 4.5% above the 28-day mean, and the last two days of the month about 1.7%.

Both are recurring calendar events that the models should capture.

3. Models

#	Model	Holidays	Events (payday, month-end, sale)
M1	Seasonal naïve: the same weekday last week	–	–
M2	SARIMAX, order chosen by AIC	✓ (dummies)	✓ (dummies)

#	Model	Holidays	Events (payday, month-end, sale)
M3	Prophet, base: trend + weekly + yearly	–	–
M4	Prophet + holidays	✓	–
M5	Prophet + holidays + events	✓	✓

M3 → M4 → M5 is the core comparison of this project: each step adds one kind of calendar knowledge. SARIMAX gets the full calendar information (the same as M5), so the Prophet vs SARIMAX comparison is fair.

3.1 Prophet

In multiplicative mode Prophet models

$$y(t) = g(t)\left[1 + s(t) + h(t) + \beta^{\top} x(t)\right] + \varepsilon_t$$

Term	Meaning
$g(t)$	piecewise-linear **trend**; its slope can change at "changepoints"
$s(t)$	**seasonality** as Fourier series (weekly, yearly)
$h(t)$	**holiday/event** effects: one coefficient per day of each window, so Prophet learns the *shape* of a festival (e.g. Diwali's ramp)
$x(t)$	extra **regressors** (payday, month-end), with coefficients β

Why these choices:

- **Multiplicative mode:** volume grew about 5×, and a festival adds a *percentage* of current volume, not a fixed count.
- **Payday and month-end as regressors:** they are known in advance, and a 30.5-day Fourier term cannot line up with calendar months.
- **A sale with a different length each year:** each year gets its own window length via `upper_window`.

Settings:

- `changepoint_prior_scale = 0.005` gives a smooth trend.
- `yearly_seasonality = 3` is a low-order yearly curve. A flexible yearly curve partly absorbs Diwali, because Diwali always falls in Oct–Nov.
- These two values performed best in a preliminary cross-validated search. The holiday and seasonality prior scales made almost no difference, so they stay at Prophet's defaults.

In [9]:
```
PROPHET_SETTINGS = dict(growth="linear", seasonality_mode="multiplicative",
changepoint_prior_scale=0.005,
                        yearly_seasonality=3, weekly_seasonality=True,
daily_seasonality=False,
                        interval_width=0.90, uncertainty_samples=500)


def make_prophet(model):
    """M3 = base, M4 = + holidays, M5 = + holidays + events."""
    holidays = {"M3": None, "M4": HOLIDAYS, "M5": HOLIDAYS_AND_EVENTS}
[model]
    m = Prophet(holidays=holidays, **PROPHET_SETTINGS)
    if model == "M5":
        m.add_regressor("is_payday", mode="multiplicative")
        m.add_regressor("is_monthend", mode="multiplicative")
    return m


FIT_COLS, PRED_COLS = ["ds", "y", "is_payday", "is_monthend"], ["ds",
"is_payday", "is_monthend"]
```

3.2 SARIMAX: design matrix and order selection by AIC

SARIMAX here is a **regression with seasonal ARIMA errors on log volume**. Its exogenous variables carry the same calendar information as Prophet M5:

- one 0/1 dummy per (holiday or sale, day of window);
- payday and month-end dummies;
- yearly Fourier terms.

`trend="c"` adds a drift term (constant growth after differencing).

The order is chosen with the course's `optimize_SARIMAX` procedure:

- fit every (p, q, P, Q) with $p, q \in \{0, 1, 2\}$, $P, Q \in \{0, 1\}$, $d = 1$, $D = 0$, $s = 7$ (from Section 2.3);
- keep the lowest **AIC**.

The search uses **only the first two years** (2022–2023), so nothing from the evaluation period influences the choice. 36 models are fitted in parallel, which takes a few minutes.

In [10]:
```
def calendar_exog(dates, table=HOLIDAYS_AND_EVENTS, fourier_k=3):
    """One dummy per (holiday/event, day offset) + payday + month-end +
yearly Fourier terms."""
    dates = pd.DatetimeIndex(dates)
    X = {}
    for _, r in table.iterrows():
        for k in range(int(r.lower_window), int(r.upper_window) + 1):
            col = f"{r.holiday}_{k:+d}"
            X.setdefault(col, np.zeros(len(dates)))
            X[col][dates == r.ds + pd.Timedelta(days=k)] = 1.0
    X = pd.DataFrame(X, index=dates)
    X["is_payday"] = (dates.day <= 5).astype(float)
```

```
    X["is_monthend"] = (dates.day > dates.days_in_month - 2).astype(float)
    t = dates.dayofyear.values / 365.25
    for k in range(1, fourier_k + 1):
        X[f"sin{k}"], X[f"cos{k}"] = np.sin(2 * np.pi * k * t), np.cos(2 *
np.pi * k * t)
    return X


X_all = calendar_exog(pd.date_range(df.ds.min(), "2027-03-31"))


def optimize_SARIMAX(endog, exog, parameters_list, d, D, s):
    """Course function (ATSA notebook 8), run in parallel. Returns (p,q,P,Q)
with AIC and BIC, best first."""
    def fit_one(param):
        quiet()
        try:
            r = SARIMAX(endog, exog, order=(param[0], d, param[1]),
seasonal_order=(param[2], D, param[3], s),
                        trend="c",
simple_differencing=False).fit(disp=False, maxiter=200)
            return [param, r.aic, r.bic, r.params]
        except Exception:
            return None
    results = [r for r in Parallel(n_jobs=N_JOBS)(delayed(fit_one)(p) for p
in parameters_list) if r]
    return (pd.DataFrame(results, columns=["(p,q,P,Q)", "AIC", "BIC",
"params"])
            .sort_values("AIC").reset_index(drop=True))


first_two_years = df[df.ds < "2024-01-01"]
endog = np.log(first_two_years.y.values)
exog = X_all.loc[first_two_years.ds]
exog = exog.loc[:, exog.sum() != 0]
t0 = time.time()
aic_tab = optimize_SARIMAX(endog, exog, list(itertools.product(range(3),
range(3), range(2), range(2))),
                           d=1, D=0, s=7)
print(f"36 models in {time.time() - t0:.0f}s")
aic_tab[["(p,q,P,Q)", "AIC", "BIC"]].head(5).round(1)
```

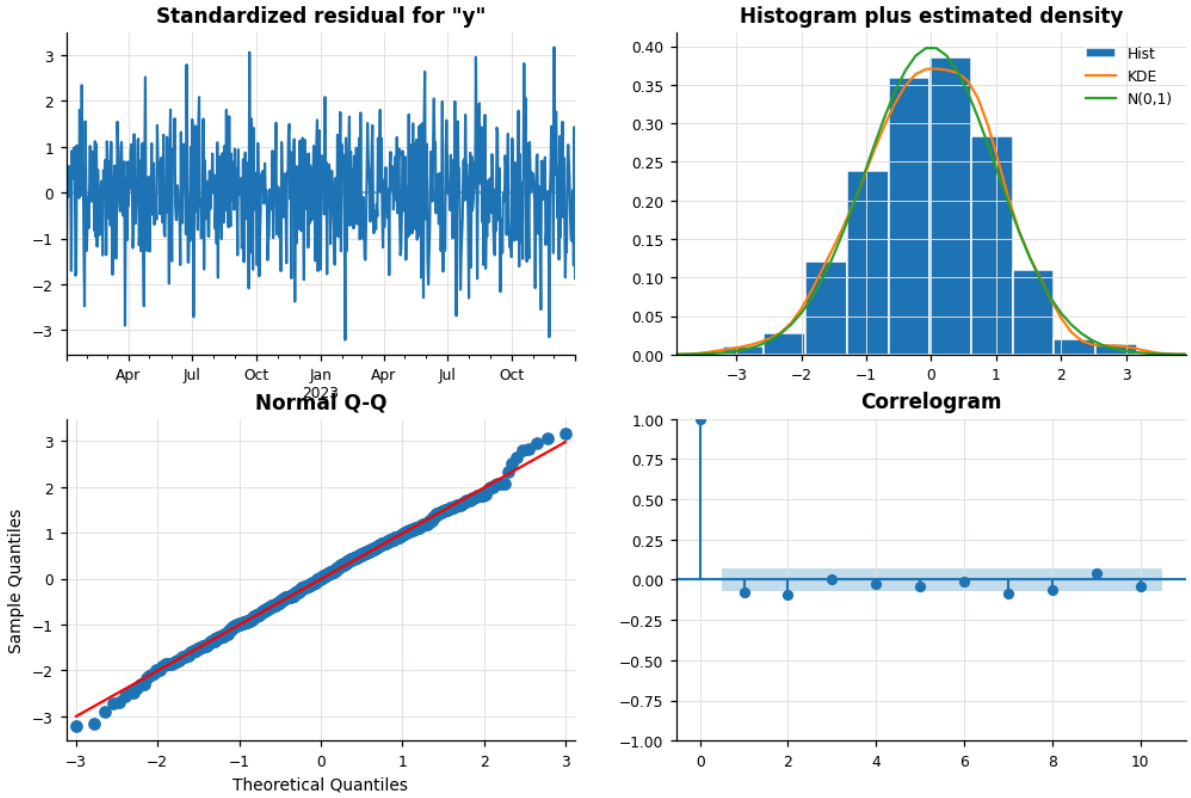
```
36 models in 149s
```

Out[10]:

	(p,q,P,Q)	AIC	BIC
0	(1, 1, 1, 1)	-3080.1	-2795.4
1	(2, 1, 1, 1)	-3073.5	-2784.3
2	(0, 2, 1, 1)	-3070.9	-2786.2
3	(1, 2, 1, 1)	-3066.5	-2777.2
4	(2, 2, 1, 1)	-3048.4	-2754.6

In [11]:
```
p, q, P, Q = aic_tab.loc[0, "(p,q,P,Q)"]
SARIMAX_ORDER = {"order": (p, 1, q), "seasonal_order": (P, 0, Q, 7)}
# Re-use the exact parameters found in the search, so this summary and AIC
match the table above
sar = SARIMAX(endog, exog, **SARIMAX_ORDER, trend="c",
simple_differencing=False).smooth(aic_tab.loc[0, "params"])
print(f"Selected:
SARIMAX{SARIMAX_ORDER['order']}x{SARIMAX_ORDER['seasonal_order']}   "
      f"AIC = {sar.aic:.1f}   BIC = {sar.bic:.1f}")
lb_sar = acorr_ljungbox(sar.resid[8:], lags=[7, 14], return_df=True)
print(f"Ljung-Box p-values on SARIMAX residuals: lag 7 =
{lb_sar.lb_pvalue.iloc[0]:.3f}, "
      f"lag 14 = {lb_sar.lb_pvalue.iloc[1]:.3f}")
fig = sar.plot_diagnostics(figsize=(11, 7))
show(fig, "3_2_sarimax_diagnostics")
```

```
Selected: SARIMAX(1, 1, 1)x(1, 0, 1, 7)   AIC = -3080.1   BIC = -2795.4
Ljung-Box p-values on SARIMAX residuals: lag 7 = 0.010, lag 14 = 0.001
```



Reading:

- **AIC selects SARIMAX(1,1,1)(1,0,1)$_7$:** one AR and one MA term after a first difference, plus a weekly seasonal AR and MA term.
- **The five best models all have $P = Q = 1$,** so the weekly seasonal terms matter most.
- **The diagnostic plots** show standardised residuals without obvious pattern and a near-normal histogram and Q-Q plot (slightly heavy tails). The correlogram is mostly inside the confidence band.

- **Ljung–Box finds a small amount of autocorrelation** (p = 0.010 at lag 7 and 0.001 at lag 14). The residuals are not perfectly white noise, but they are far cleaner than Prophet's ($p \approx 10^{-88}$, Section 6).
- **So by the course's in-sample criteria this is a reasonable, if imperfect, model.** Section 4 tests whether it also forecasts well.

4. Forecast evaluation

4.1 Rolling-origin cross-validation

A random train/test split would let a model see the future, so we use **rolling-origin evaluation** instead:

- Train on all data up to a cutoff and forecast the next **90 days**.
- Move the cutoff forward **90 days** and repeat.
- The first cutoff is after **2 years** of data, so yearly patterns can be learned.

This gives **10 folds** whose forecasts cover Jan 2024 – Aug 2026, and **every model uses exactly the same folds**. All errors below are measured on data the model had not seen. That is stricter than scoring a model on the data it was fitted to.

Metrics:

- **MAPE**: mean absolute percentage error; the main metric.
- **MAE**: mean absolute error, in million transactions per day.
- **RMSE**: like MAE, but penalises large misses more.

In [12]:
```
HORIZON, INITIAL, STEP = 90, 730, 90
EVAL_HORIZONS = (7, 30, 90)  # compare short-, medium-, and longer-horizon
performance
cutoffs = list(pd.date_range(df.ds.min() + pd.Timedelta(days=INITIAL),
                             df.ds.max() - pd.Timedelta(days=HORIZON),
freq=f"{STEP}D"))


def forecast_fold(model, cutoff):
    """Train `model` on data up to `cutoff` and forecast the next HORIZON
days."""
    quiet()
    train = df[df.ds ⩽ cutoff]
    test = df[(df.ds > cutoff) & (df.ds ⩽ cutoff +
pd.Timedelta(days=HORIZON))]
    out = test[["ds", "y", "day_type"]].copy()
    out["model"], out["cutoff"] = model, cutoff
    if model == "M1":                                          # seasonal
naive
        last_week = train.set_index("ds").y.iloc[-7:]
        out["yhat"] = test.ds.dt.dayofweek.map({d.dayofweek: v for d, v in
last_week.items()}).values
```

```
    elif model == "M2":                                          # SARIMAX on
log volume
        Xtr = X_all.loc[train.ds]
        Xtr = Xtr.loc[:, Xtr.sum() != 0]                         # drop dummies
for events not yet seen
        res = SARIMAX(np.log(train.y.values), Xtr.values, **SARIMAX_ORDER,
trend="c").fit(disp=False, maxiter=500)
        pred = res.get_forecast(len(test), exog=X_all.loc[test.ds,
Xtr.columns].values)
        # Lognormal mean correction: exp(mu + 0.5 * sigma^2) approximates
the conditional mean
        # when log-scale forecast errors are approximately normal. Use
training residual variance
        # as a simple, transparent correction; assess it empirically with
the held-out metrics.
        sigma2 = float(np.nanvar(np.asarray(res.resid)[1:], ddof=1))
        out["yhat"] = np.exp(np.asarray(pred.predicted_mean) + 0.5 *
max(sigma2, 0.0))
    else:                                                        # Prophet M3 /
M4 / M5
        m = make_prophet(model).fit(train[FIT_COLS])
        out["yhat"] = m.predict(test[PRED_COLS]).yhat.values
    out["yhat"] = np.maximum(out["yhat"].to_numpy(), 0.0)
    out["lead_days"] = (out["ds"] - cutoff).dt.days
    return out


def mape(d): return 100 * np.mean(np.abs(d.y - d.yhat) / d.y)
def mae(d):  return np.mean(np.abs(d.y - d.yhat))
def rmse(d): return np.sqrt(np.mean((d.y - d.yhat) ** 2))
def wape(d): return 100 * np.sum(np.abs(d.y - d.yhat)) / np.sum(np.abs(d.y))
def smape(d):
    denom = np.abs(d.y) + np.abs(d.yhat)
    return 200 * np.mean(np.abs(d.y - d.yhat) / np.where(denom == 0, np.nan,
denom))
def mean_error(d): return np.mean(d.yhat - d.y)


LABELS = {"M1": "M1 Seasonal naïve", "M2": "M2 SARIMAX", "M3": "M3 Prophet
base",
          "M4": "M4 Prophet + holidays", "M5": "M5 Prophet + holidays +
events"}
t0 = time.time()
cv = pd.concat(Parallel(n_jobs=N_JOBS)(delayed(forecast_fold)(m, c) for m in
LABELS for c in cutoffs),
               ignore_index=True)
cv.to_csv(OUT / "cv_forecasts.csv", index=False)
print(f"{len(LABELS)} models × {len(cutoffs)} folds in {time.time() -
t0:.0f}s")
```

```
5 models × 10 folds in 103s
```

4.2 Results

In [13]:
```
rows = []
for m, label in LABELS.items():
    g = cv[cv.model == m]
    rows.append({"model": label, "MAPE %": mape(g), "WAPE %": wape(g),
"sMAPE %": smape(g),
                 "MAE (million/day)": mae(g), "RMSE (million/day)":
rmse(g),
                 "Mean error (million/day)": mean_error(g),
                 **{f"MAPE {t} %": mape(g[g.day_type == t]) for t in
DAY_TYPES}})
results = pd.DataFrame(rows)
results.to_csv(OUT / "results_table.csv", index=False)
print(f"SARIMAX (selection window): AIC = {sar.aic:.1f}, BIC =
{sar.bic:.1f}")
results.round(2)
```

```
SARIMAX (selection window): AIC = -3080.1, BIC = -2795.4
```

Out[13]: Columns 1–8 of 11

	model	MAPE %	WAPE %	sMAPE %	MAE (million/day)	RMSE (million/day)	Mean error (million/day)
0	M1 Seasonal naïve	5.54	5.56	5.77	32.48	41.08	-26.86
1	M2 SARIMAX	4.73	4.90	4.63	28.64	38.03	12.31
2	M3 Prophet base	3.32	3.33	3.35	19.48	25.43	-4.11
3	M4 Prophet + holidays	3.17	3.17	3.20	18.55	23.87	-3.15
4	M5 Prophet + holidays + events	2.44	2.43	2.44	14.20	18.14	-0.71

Columns 9–11 of 11 · first column repeated

	MAPE ordinary %	MAPE holiday window %	MAPE sale event %
0	5.29	7.88	11.01
1	4.61	6.32	4.56
2	3.11	5.21	7.26
3	3.07	3.64	8.56
4	2.36	3.10	4.12

4.3 Performance by forecast horizon and fold

Overlapping rolling-origin folds are useful for evaluating forecast operations, but their errors are not independent observations. We therefore report pooled errors by lead time and the spread in

fold-level MAPE, rather than treating every forecasted day as an independent replicate. WAPE complements MAPE by weighting errors according to transaction volume.

In [14]:

```
# Pooled error by forecast lead. lead_days is computed from each fold's
forecast origin.
if "lead_days" not in cv.columns:
    raise ValueError("lead_days is missing; rerun the forecast-evaluation
cells above.")

horizon_rows = []
for model, label in LABELS.items():
    g = cv[cv.model == model]
    for h in EVAL_HORIZONS:
        gh = g[g.lead_days <= h]
        if gh.empty:
            continue
        horizon_rows.append({
            "model": label,
            "horizon_days": h,
            "MAPE %": mape(gh),
            "WAPE %": wape(gh),
            "MAE (million/day)": mae(gh),
            "RMSE (million/day)": rmse(gh),
        })
horizon_results = pd.DataFrame(horizon_rows)
horizon_results.to_csv(OUT / "horizon_results.csv", index=False)
horizon_results.round(3)

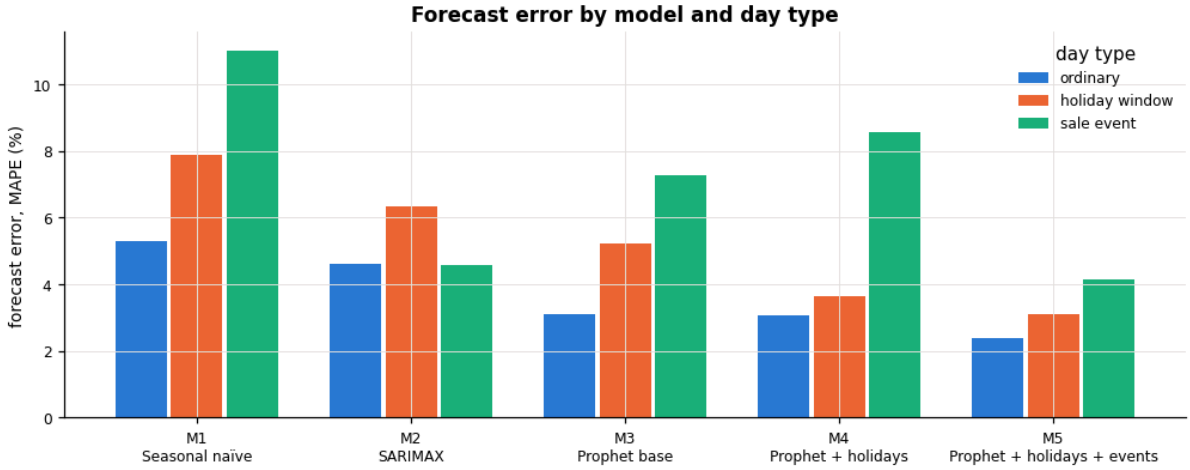
# Fold-level MAPE variation. These folds overlap in time, so this is
descriptive variability,
# not a formal independent-sample confidence interval.
fold_rows = []
for (model, cutoff), g in cv.groupby(["model", "cutoff"]):
    fold_rows.append({"model": LABELS[model], "cutoff": cutoff,
                      "MAPE %": mape(g), "WAPE %": wape(g), "MAE": mae(g),
"RMSE": rmse(g)})
fold_metrics = pd.DataFrame(fold_rows)
fold_summary = (fold_metrics.groupby("model")["MAPE %"]
                .agg(["mean", "std", "min", "max"]).rename(columns={"mean":
"mean fold MAPE %",
                "std": "fold MAPE SD", "min": "best fold MAPE %", "max":
"worst fold MAPE %"}))
fold_metrics.to_csv(OUT / "fold_metrics.csv", index=False)
fold_summary.to_csv(OUT / "fold_summary.csv")
fold_summary.round(3)
```

Out[14]:

model	mean fold MAPE %	fold MAPE SD	best fold MAPE %	worst fold MAPE %
M1 Seasonal naïve	5.543	1.120	4.026	7.261
M2 SARIMAX	4.733	2.095	2.572	8.729
M3 Prophet base	3.316	0.449	2.886	4.328
M4 Prophet + holidays	3.172	0.421	2.837	4.298
M5 Prophet + holidays + events	2.438	0.354	1.914	3.300

In [15]:

```
fig, ax = plt.subplots(figsize=(11, 3.8))
w = 0.26
for i, (t, col) in enumerate(zip(DAY_TYPES, [C_ACTUAL, C_MODEL, C_ALT])):
    ax.bar(np.arange(len(LABELS)) + (i - 1) * w, results[f"MAPE {t} %"],
width=w - 0.02, color=col, label=t)
ax.set_xticks(range(len(LABELS)), [l.replace(" ", "\n", 1) for l in
LABELS.values()], fontsize=8)
ax.set_ylabel("forecast error, MAPE (%)")
ax.legend(title="day type")
ax.set_title("Forecast error by model and day type")
show(fig, "4_2_error_by_day_type")
```



Reading:

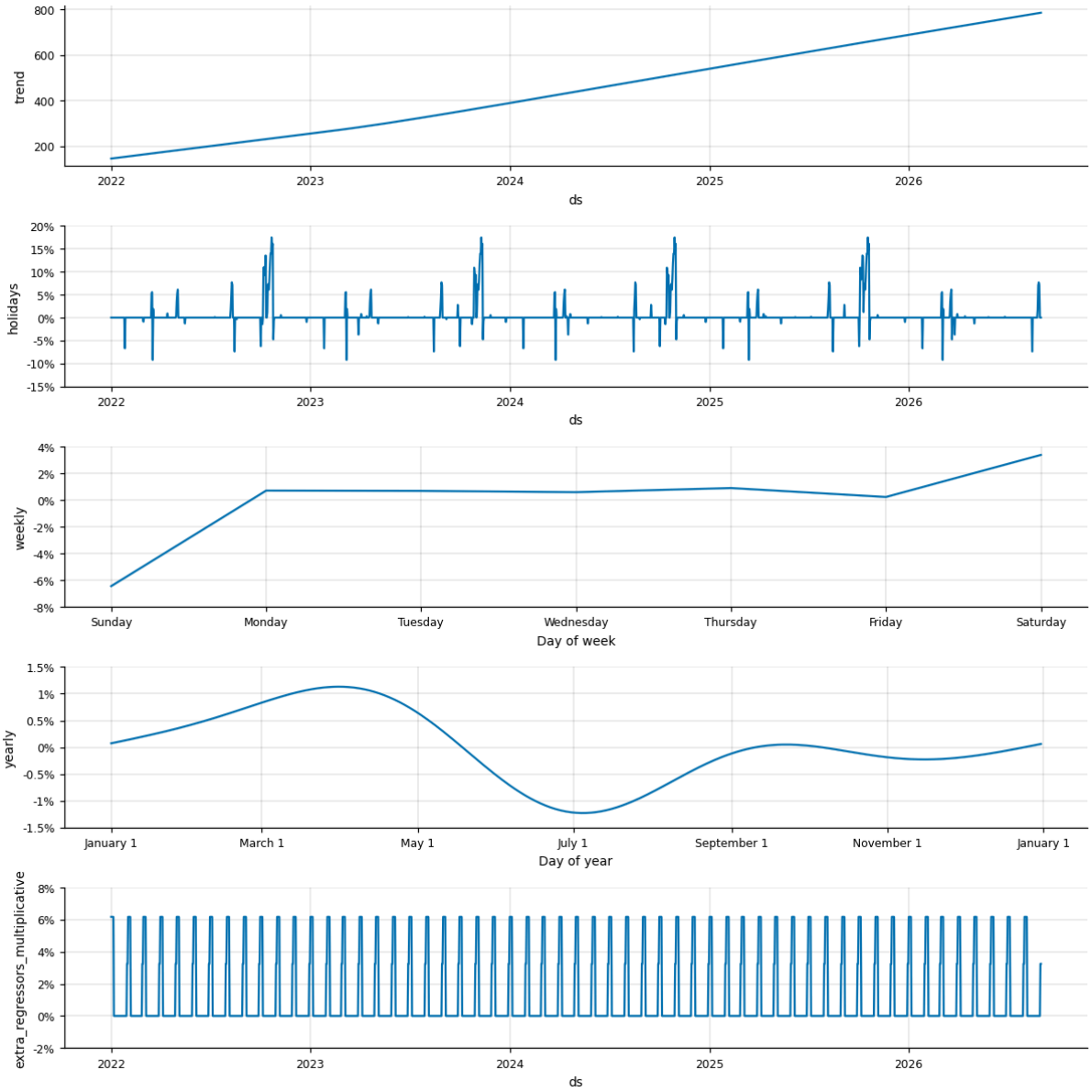
- **Prophet with holidays and events (M5) is the most accurate model:** 2.44% MAPE, less than half the seasonal naïve error (5.54%).
- **Holidays help on holiday days.** Adding them (M3 → M4) cuts holiday-window error from 5.21% to 3.64%, about 30% lower, while ordinary days barely change (3.11% → 3.07%).
- **Events give the biggest overall gain** (M4 → M5: 3.17% → 2.44%):
  - Payday and month-end affect seven days of every month, so ordinary-day error falls from 3.07% to 2.36%.
  - The sale's own days improve from 8.56% to 4.12%. Without the sale in the model, those days are under-forecast by about 8.6% on average.

- **SARIMAX, given the same calendar information, reaches only 4.73%.** It over-forecasts holiday days by about 5.3% on average, and its errors build up over the horizon: 2.92% at 7 days, 3.25% at 30 days and 4.73% at 90 days, against 2.15%, 2.21% and 2.44% for M5. Its fold-to-fold spread is also the widest (MAPE from 2.6% to 8.7%). Prophet's structure (deterministic trend plus calendar terms) suits long-horizon daily forecasting better.
- **A good fit is not a good forecast.** SARIMAX's in-sample residuals were much closer to white noise than Prophet's, yet it forecast worse than every Prophet variant. That is why we judge all models on out-of-sample forecasts.

5. Holiday and event effects

We fit the full model (M5) on all the data and read off its estimated effects. Each holiday or event column in Prophet's output is the multiplicative effect on that day (0.12 means +12%). We compare these with the true injected effects, which is possible only because the data is synthetic.

In [16]:
```
m5 = make_prophet("M5").fit(df[FIT_COLS])
f5 = m5.predict(df[PRED_COLS])
fig = m5.plot_components(f5, figsize=(11, 11))
show(fig, "5_components")
```



Reading the components:

- **trend:** smooth growth.
- **holidays:** spikes and dips at each festival and sale.
- **weekly:** Saturday high, Sunday low.
- **yearly:** a gentle annual cycle.
- **extra_regressors_multiplicative:** the payday and month-end lift.

5.1 Estimated against true effect for each holiday and event

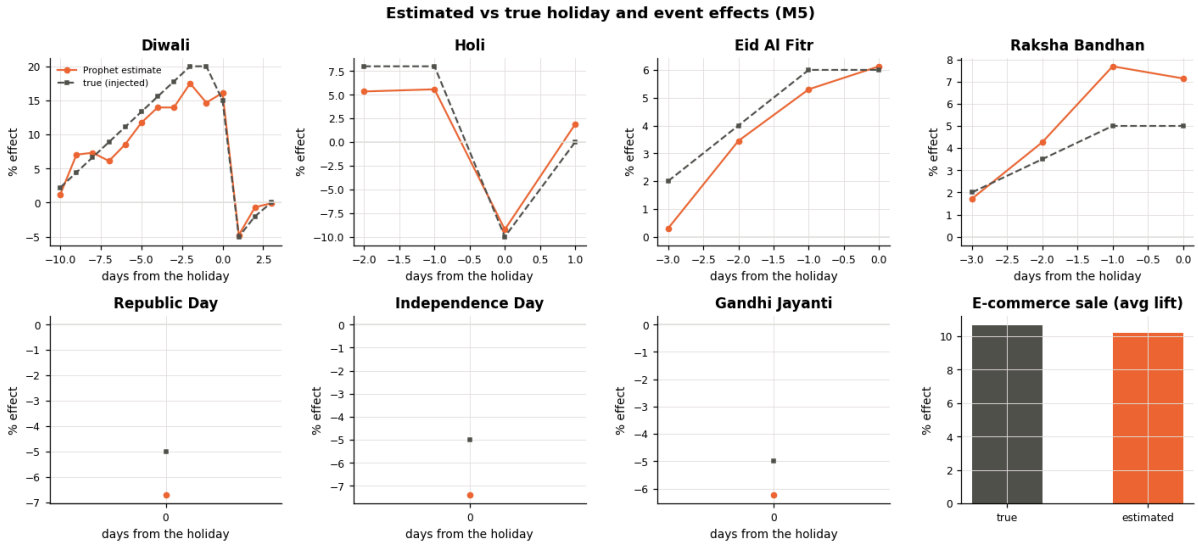
In [17]:
```
offsets = pd.DataFrame([(r.holiday, k, r.ds + pd.Timedelta(days=k))
                        for _, r in pd.concat([festivals, sale]).iterrows()
                        for k in range(int(r.lower_window),
int(r.upper_window) + 1)],
                       columns=["name", "offset", "date"])
offsets = offsets[offsets.date ⩽ df.ds.max()]
f5i, dfi = f5.set_index("ds"), df.set_index("ds")
offsets["estimated"] = [f5i.loc[d, n] for n, d in zip(offsets.name,
offsets.date)]
offsets["true"] = dfi.loc[offsets.date, "true_effect"].values
offsets.loc[offsets.name == "ecommerce_sale", "offset"] = 0   # sale lengths
differ by year; compare its average lift

effects = (offsets.groupby("name", sort=False).agg(days_in_window=("offset",
"nunique"),
                                                   true=("true", "mean"),
estimated=("estimated", "mean")))
effects[["true", "estimated"]] *= 100
effects["error (pp)"] = effects.estimated - effects.true
calendar = pd.DataFrame({
    "true": [100 * truth["payday"]["lift"], 100 * truth["monthend"]
["lift"]],
    "estimated": [100 * f5.loc[df.is_payday == 1, "is_payday"].mean(),
                  100 * f5.loc[df.is_monthend == 1, "is_monthend"].mean()]},
    index=["payday (days 1-5)", "month-end (last 2 days)"])
calendar["error (pp)"] = calendar.estimated - calendar.true
effect_table = pd.concat([effects.drop(columns="days_in_window"), calendar])
effect_table.index.name = "effect (mean over its window, %)"
effect_table.to_csv(OUT / "effect_recovery.csv")
effect_table.round(2)
```

Out[17]:

effect (mean over its window, %)	true	estimated	error (pp)
diwali	9.14	8.05	-1.10
holi	1.50	0.89	-0.61
eid_al_fitr	4.50	3.79	-0.71
raksha_bandhan	3.87	5.20	1.33
republic_day	-5.00	-6.70	-1.70
independence_day	-5.00	-7.40	-2.40
gandhi_jayanti	-5.00	-6.23	-1.23
ecommerce_sale	10.67	10.21	-0.46
payday (days 1–5)	6.00	6.18	0.18
month-end (last 2 days)	3.00	3.25	0.25

In [18]:
```
names = list(dict.fromkeys(festivals.holiday))
fig, axes = plt.subplots(2, 4, figsize=(13, 6))
for ax, n in zip(axes.flat, names):
    g = offsets[offsets.name == n].groupby("offset")[["estimated",
"true"]].mean() * 100
    ax.plot(g.index, g.estimated, color=C_MODEL, marker="o", ms=4,
label="Prophet estimate")
    ax.plot(g.index, g.true, color=C_TRUTH, ls="--", marker="s", ms=3,
label="true (injected)")
    ax.axhline(0, color=C_GRID, lw=1)
    ax.set_title(n.replace("_", " ").title())
    ax.set_xlabel("days from the holiday")
    ax.set_ylabel("% effect")
    if len(g) == 1:
        ax.set_xlim(-1, 1)
        ax.set_xticks([0])
g = offsets[offsets.name == "ecommerce_sale"]
ax = axes.flat[-1]
ax.bar([0, 1], [100 * g.true.mean(), 100 * g.estimated.mean()], color=
[C_TRUTH, C_MODEL], width=0.5)
ax.set_xticks([0, 1], ["true", "estimated"])
ax.set_title("E-commerce sale (avg lift)")
ax.set_ylabel("% effect")
axes.flat[0].legend(fontsize=7, loc="upper left")
fig.suptitle("Estimated vs true holiday and event effects (M5)",
fontweight="bold")
fig.tight_layout()
show(fig, "5_1_effect_recovery")
```



Reading:

- **Events are recovered almost exactly:**
  - payday +6.2% (true +6.0%);
  - month-end +3.3% (true +3.0%);
  - the sale +10.2% (true +10.7% on average). The sale's real lift varies between 8% and 12% from year to year, while the model fits one average.

- **Diwali's shape is recovered:** the ramp over ten days, the peak on Dhanteras and Diwali eve, and the dip after. **Its size is under-estimated:** an 8.1% window average against a true 9.1%.
  - Each of its 14 daily coefficients is estimated from only four Diwalis.
  - Part of the festive lift is absorbed by the yearly seasonality, since Diwali always falls in Oct–Nov.
- **Holi, Eid al-Fitr and Raksha Bandhan are within about 1.3 pp.** Holi's −10% on the day itself is caught almost exactly.
- **Single-day national holidays come out 1.2 to 2.4 pp more negative than −5%.** Each occurs only four or five times, so their estimates are the noisiest.

Overall: the bigger and more frequent the effect, the more precisely Prophet recovers it.

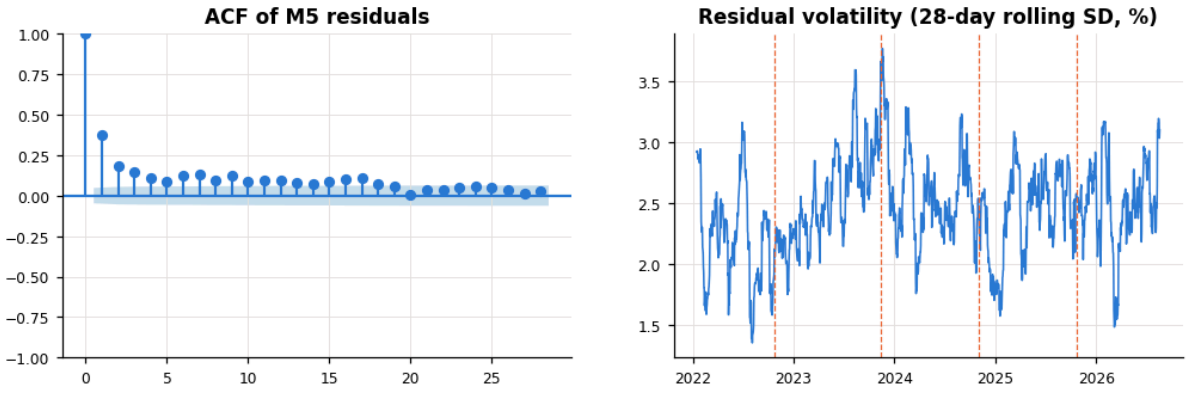
6. Residual diagnostics

A good model leaves residuals that look like random noise. We check two things on the in-sample residuals of M5 (log scale):

- **Ljung–Box:** are the residuals autocorrelated? A small p-value means yes.
- **ARCH-LM:** does the size of the residuals vary over time (heteroskedasticity)? A small p-value means yes.

In [19]:
```
res = np.log(df.y.values) - np.log(f5.yhat.values)
lb = acorr_ljungbox(res, lags=[7, 14], return_df=True)
lm, lm_p, _, _ = het_arch(res, nlags=7)
print(f"Lag-1 autocorrelation: {acf(res, nlags=1)[1]:.2f}   (the generator's
noise has 0.3)")
print(f"Ljung-Box p-values: lag 7 = {lb.lb_pvalue.iloc[0]:.2g}, lag 14 =
{lb.lb_pvalue.iloc[1]:.2g}")
print(f"ARCH-LM (7 lags): LM = {lm:.1f}, p = {lm_p:.2g}")
fig, ax = plt.subplots(1, 2, figsize=(11, 3.2))
plot_acf(res, lags=28, ax=ax[0], color=C_ACTUAL, vlines_kwargs={"colors":
C_ACTUAL})
ax[0].set_title("ACF of M5 residuals")
ax[1].plot(df.ds, pd.Series(100 * res).rolling(28, center=True).std(),
color=C_ACTUAL, lw=1)
for d in festivals.query("holiday == 'diwali'").ds:
    if d <= df.ds.max():
        ax[1].axvline(d, color=C_MODEL, lw=0.8, ls="--")
ax[1].set_title("Residual volatility (28-day rolling SD, %)")
show(fig, "6_residual_diagnostics")
```

```
Lag-1 autocorrelation: 0.38   (the generator's noise has 0.3)
Ljung-Box p-values: lag 7 = 4.1e-88, lag 14 = 4.4e-104
ARCH-LM (7 lags): LM = 29.4, p = 0.00012
```



Reading:

- **The residuals are autocorrelated.** Lag-1 is 0.38, above the 0.3 built into the noise, and Ljung–Box rejects strongly. Prophet assumes independent errors and does not model this, so it leaves short-term information unused.
- **The residual variance is not constant.** ARCH-LM gives $p \approx 1 \times 10^{-4}$, and the rolling SD changes visibly over time. The effect is mild, but Prophet's prediction intervals assume constant variance, so they are too narrow in volatile periods.
- **Both point to the same extensions:** model the residuals with ARIMA (a Prophet + ARIMA hybrid), or the variance with GARCH.

7. Final forecast: September 2026 – February 2027

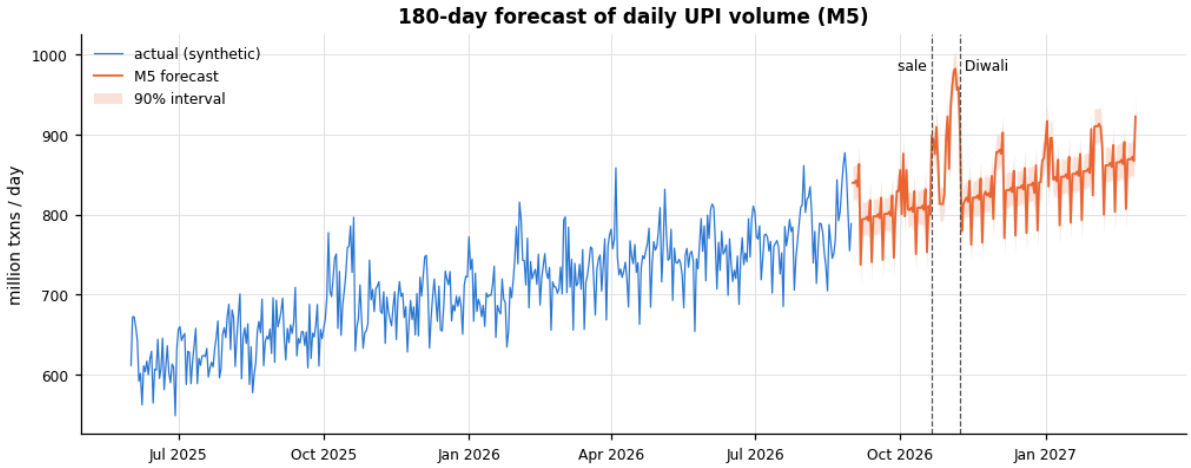
M5 is refitted on all the data and forecasts 180 days ahead:

- Payday and month-end come straight from the calendar.
- Diwali 2026 (8 November) and the 2026 sale (21–25 October) are in the holiday/event table, so both appear in the forecast.

In [20]:
```
future = m5.make_future_dataframe(periods=180)
future["is_payday"] = (future.ds.dt.day ⩽ 5).astype(int)
future["is_monthend"] = (future.ds.dt.day > future.ds.dt.days_in_month -
2).astype(int)
fc = m5.predict(future)
fc = fc[fc.ds > df.ds.max()]
fc[["ds", "yhat", "yhat_lower", "yhat_upper"]].to_csv(OUT /
"forecast_180_days.csv", index=False)

recent = df[df.ds ⩾ "2025-06-01"]
fig, ax = plt.subplots(figsize=(11, 4))
ax.plot(recent.ds, recent.y, color=C_ACTUAL, lw=0.8, label="actual
(synthetic)")
ax.plot(fc.ds, fc.yhat, color=C_MODEL, lw=1.2, label="M5 forecast")
ax.fill_between(fc.ds, fc.yhat_lower, fc.yhat_upper, color=C_MODEL,
alpha=0.18, lw=0, label="90% interval")
for d, txt, side in [("2026-10-21", "sale ", "right"), ("2026-11-08", "
Diwali", "left")]:
    ax.axvline(pd.Timestamp(d), color=C_TRUTH, ls="--", lw=0.8)
    ax.text(pd.Timestamp(d), ax.get_ylim()[1] * 0.97, txt, fontsize=8,
va="top", ha=side)
ax.xaxis.set_major_formatter(mdates.DateFormatter("%b %Y"))
ax.set_ylabel("million txns / day")
ax.legend(loc="upper left")
ax.set_title("180-day forecast of daily UPI volume (M5)")
show(fig, "7_forecast")

monthly_fc = fc.groupby(fc.ds.dt.to_period("M")).yhat.agg(["sum", "size"])
monthly_fc = pd.DataFrame({"forecast (billion txns)": (monthly_fc["sum"] /
1000).round(2),
                           "days covered": monthly_fc["size"]})
monthly_fc.index = monthly_fc.index.strftime("%b %Y")
monthly_fc
```



Out[20]:

ds	forecast (billion txns)	days covered
Sep 2026	24.08	30
Oct 2026	25.75	31
Nov 2026	25.51	30
Dec 2026	26.01	31
Jan 2027	26.44	31
Feb 2027	23.53	27

Reading:

- **The forecast continues the trend** at about 24–26 billion transactions a month (August 2026 actual: 24.5 billion). February 2027 is lower only because the forecast ends on 27 February.
- **The sale and the Diwali ramp produce the highest forecast days,** close to 1 billion transactions a day in late October and early November.
- **The 90% interval is narrow and should be read as optimistic,** given the residual autocorrelation in Section 6.

This forecasts the synthetic series; it illustrates the method and is not a prediction of real UPI volumes.

8. Conclusions

Answers to the questions

1. Do holidays and events improve the forecast, and where?

- Yes. Adding holidays cut holiday-window error by about 30% (5.21% → 3.64%).
- Adding events (payday, month-end, the sale) cut overall error from 3.17% to 2.44% and more than halved the error on sale days (8.56% → 4.12%).
- Each improvement appears exactly on the days the added information describes. Ordinary days change only when payday is added.

2. Does Prophet recover the true effects?

- Events and short holidays, closely: payday 6.2% against 6.0%, the sale 10.2% against 10.7%, Holi and Eid within about 1 pp.
- Diwali's shape is right, but its size is under-estimated (8.1% against 9.1%). It is a long festival seen only four times, and it overlaps the yearly cycle.

3. Prophet vs SARIMAX?

- With the same calendar information, Prophet M5 (2.44%) clearly beat SARIMAX (4.73%), whose errors grow with the forecast horizon. M5 was also more accurate at every horizon (7, 30 and 90 days).

- SARIMAX had much cleaner in-sample residuals, which shows that a good fit does not guarantee a good forecast.

Limitations

The reported metrics are specific to this synthetic data generator and the chosen forecast origins. They should not be presented as validated real-world UPI forecast accuracy. Forecast-fold overlap also means fold scores are descriptive and not independent replicates.

- **The daily data is synthetic.** The effect sizes are our assumptions, so the results show that Prophet can recover effects of this kind, not the real size of Diwali's effect on UPI.
- **Five festive months deviate from NPCI totals** by up to 6.5%, by design (`DATA.md` §3.4).
- **Only four Diwalis** are in the sample.
- **Prophet's residuals are autocorrelated and slightly heteroskedastic,** so its prediction intervals are too narrow.

Future scope

- Use real daily payment data if NPCI or RBI publish it.
- Build a Prophet + ARIMA hybrid for the residual autocorrelation, or GARCH for the changing variance.
- Detect unannounced events automatically from forecast residuals.
- Compare with machine-learning models (e.g. LSTM, gradient boosting on calendar features).
- Model P2P and P2M payments together (VAR/VARMAX) to see how festivals shift the mix.

Reproducibility checklist

- Keep the `data/` directory beside this notebook with `synthetic_daily.csv`, `upi_monthly.csv`, `prophet_holidays.csv`, `holidays.csv`, and `truth.json`.
- Keep `DATA.md` with the documented NPCI source URLs and the synthetic data-generation assumptions.
- Install compatible versions of `prophet`, `statsmodels`, `pandas`, `numpy`, `matplotlib`, and `joblib`.
- Run cells from top to bottom. Outputs and CSV summaries are saved under `results/`.
- The notebook validates the daily index and input columns before modelling. The SARIMAX lognormal correction is an approximation.
- Synthetic event effects are known by construction; effect-recovery tables are a simulation check, not causal estimates for real UPI.